Software Requirements Specification

for

AI-Based Email Spam Detection

Prepared by: 1. Aashi Kothale

44. Shashwat Bujone

51. Yogeshwar Kinekar

Ramdeobaba University

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1. Introduction

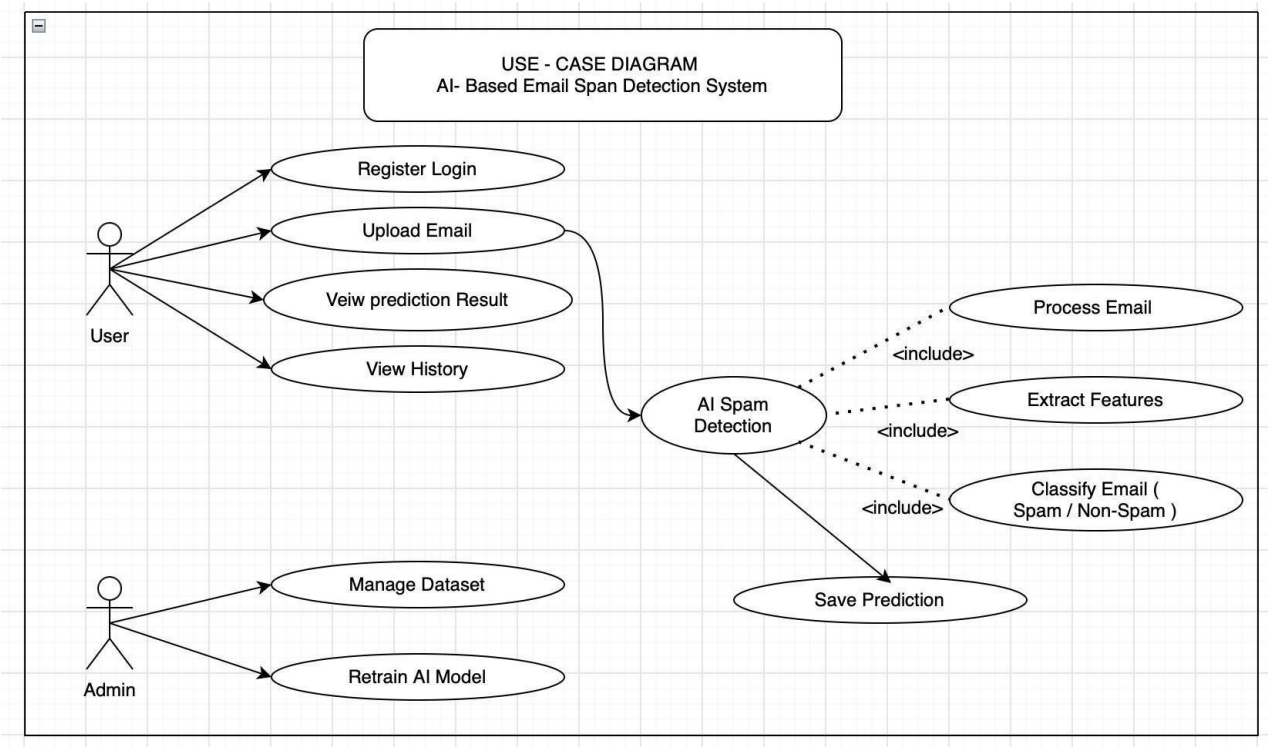
1.1 Introduction

Email has become one of the most widely used methods of communication for individuals and organizations. However, the increasing volume of unwanted and malicious emails, commonly known as spam, has created significant security and productivity challenges. Spam emails may contain advertisements, fraudulent offers, phishing links, malware, or other harmful content that can compromise users and their data.

*The **AI-Based Email Spam Detection System** is designed to automatically identify and classify incoming emails as either **spam or legitimate (ham)** using Artificial Intelligence and Machine Learning techniques. The system analyses various characteristics of an email, such as its subject, message content, keywords, links, and other relevant patterns, to determine whether the email is suspicious.*

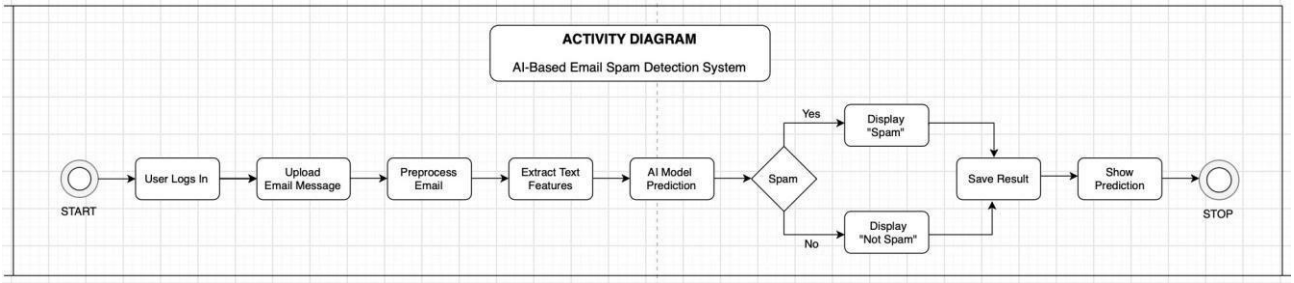
By using machine learning algorithms, the proposed system can learn from previously classified emails and improve its ability to detect new and evolving forms of spam. This reduces the need for manual email filtering and helps users save time while improving security. The system aims to provide an efficient, accurate, and automated solution for detecting spam emails and protecting users from potentially harmful messages.

2. Use – Case Diagram

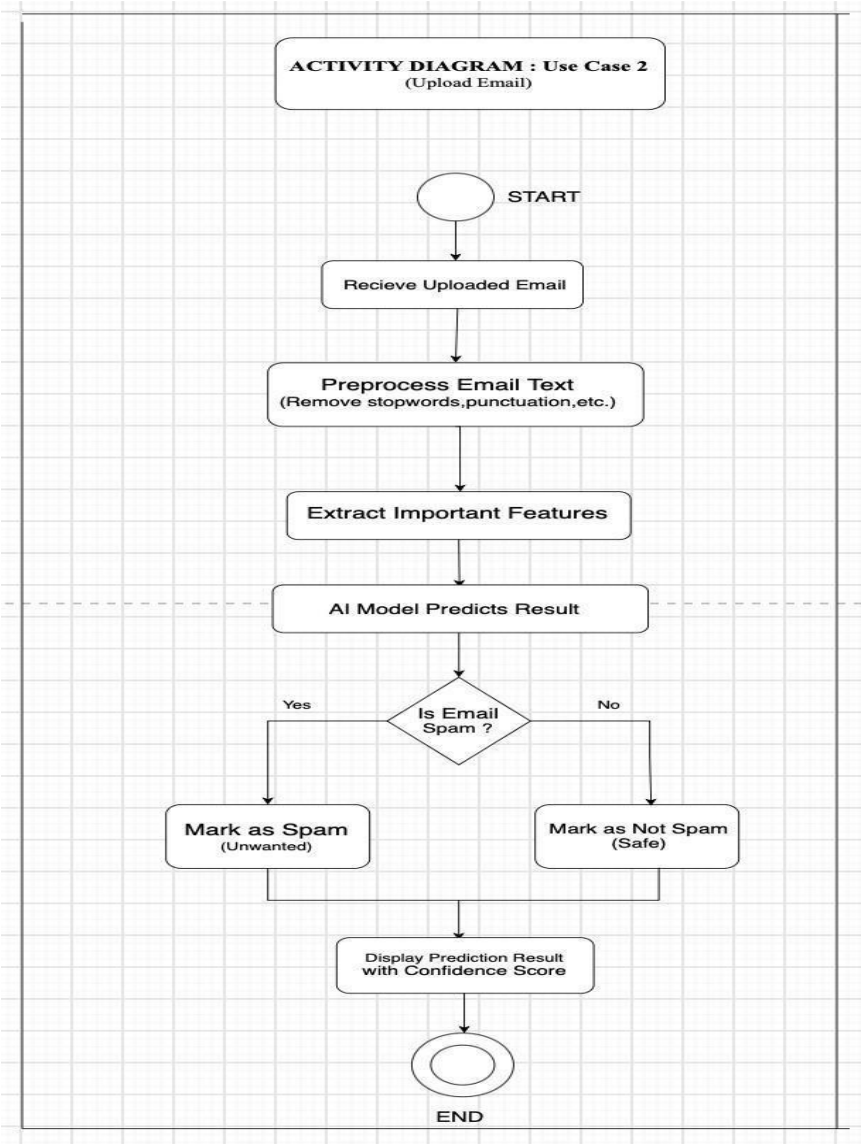


3. Activity Diagram

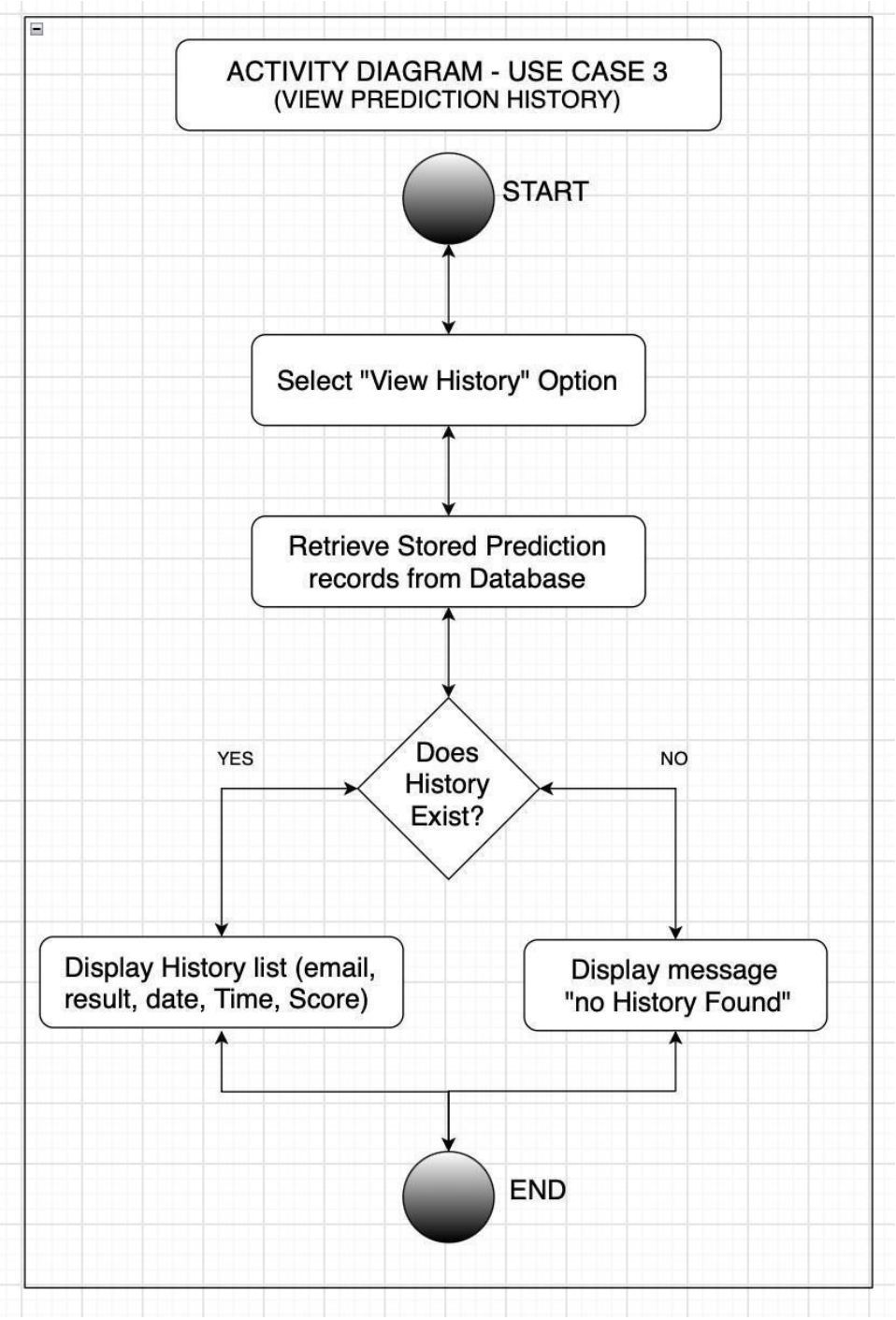
3.1 Overall Activity Diagram



3.2 Activity Diagram Case – II

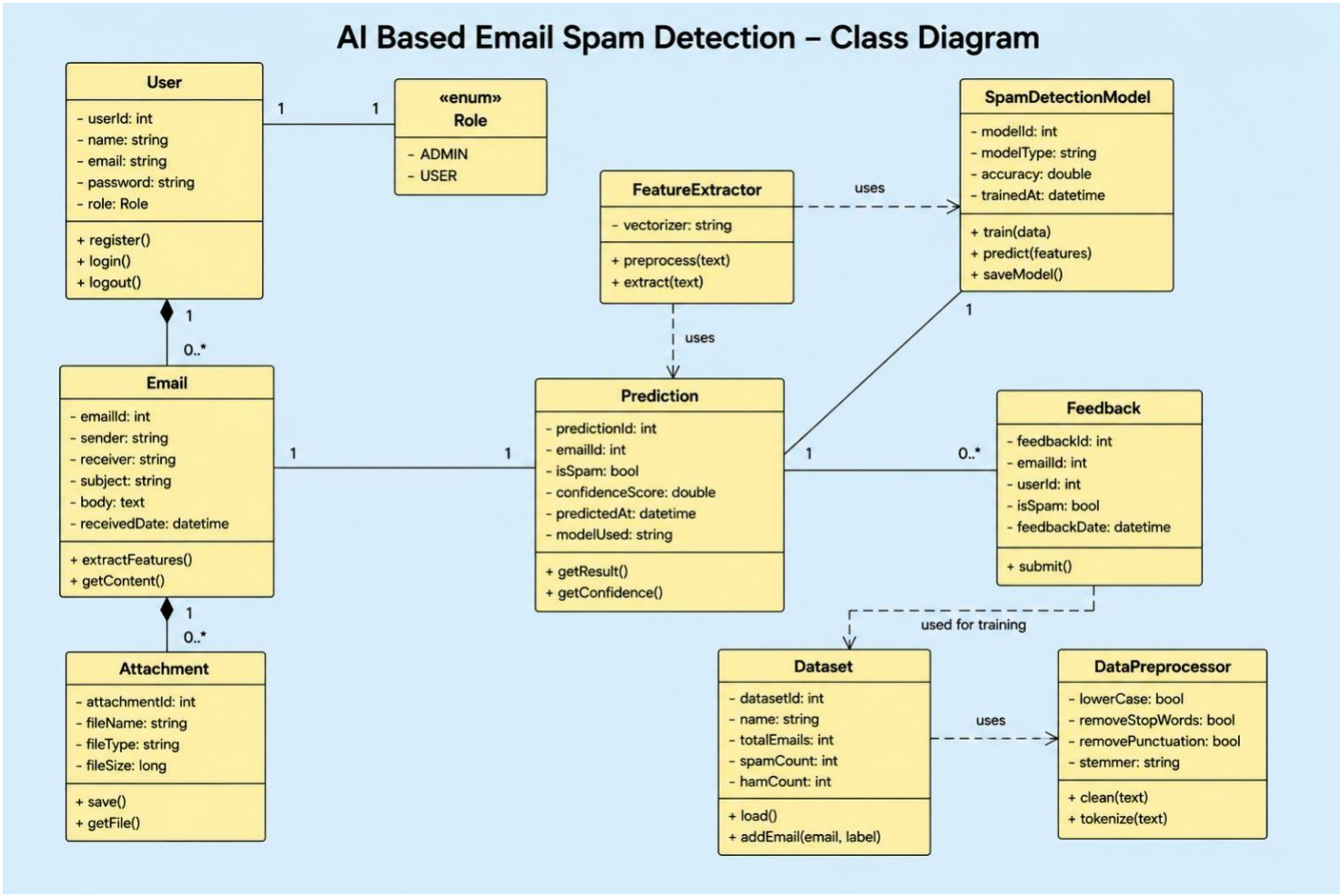


3.3 Activity Diagram Case – III



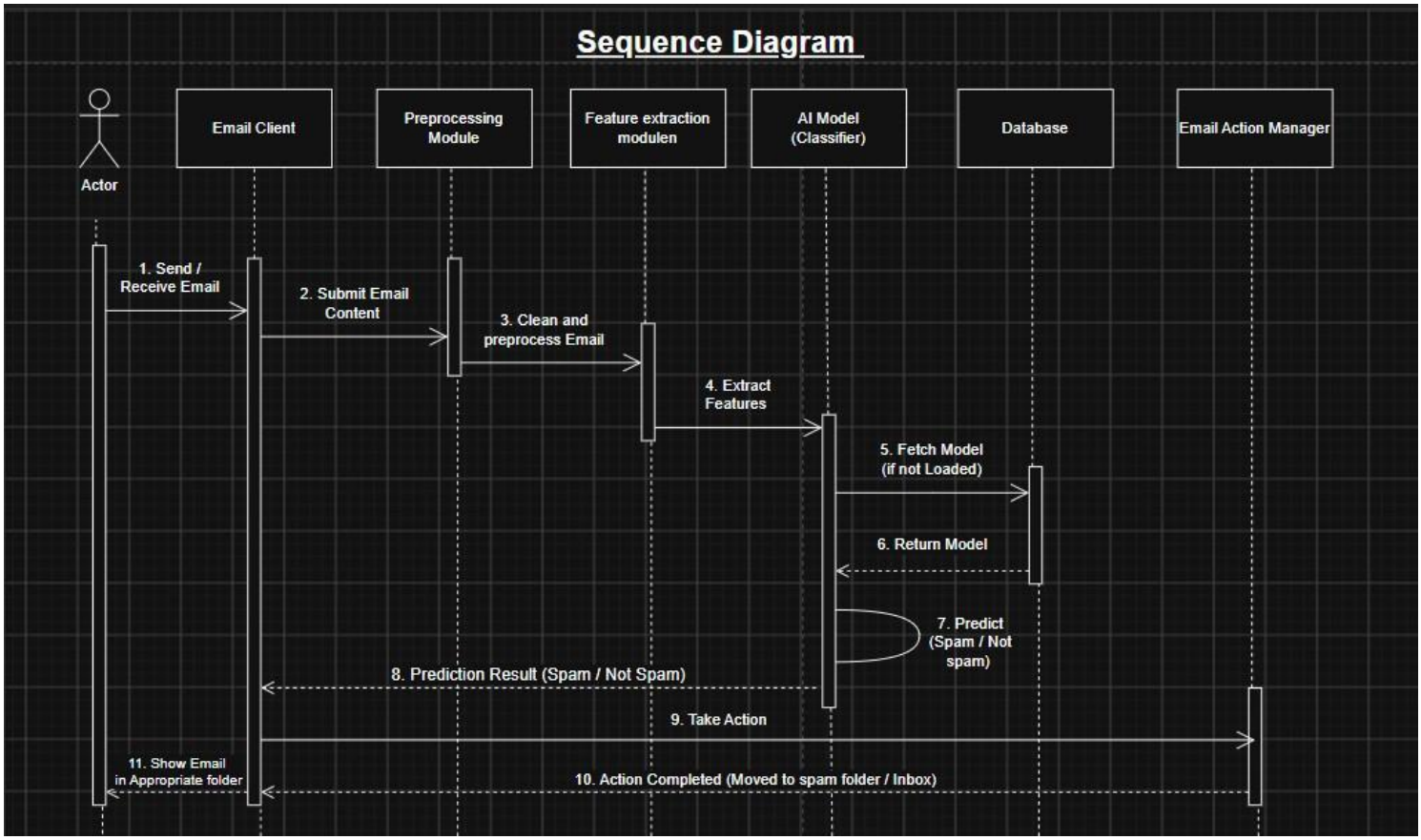
4. Class Diagram

4.1 Class Diagram



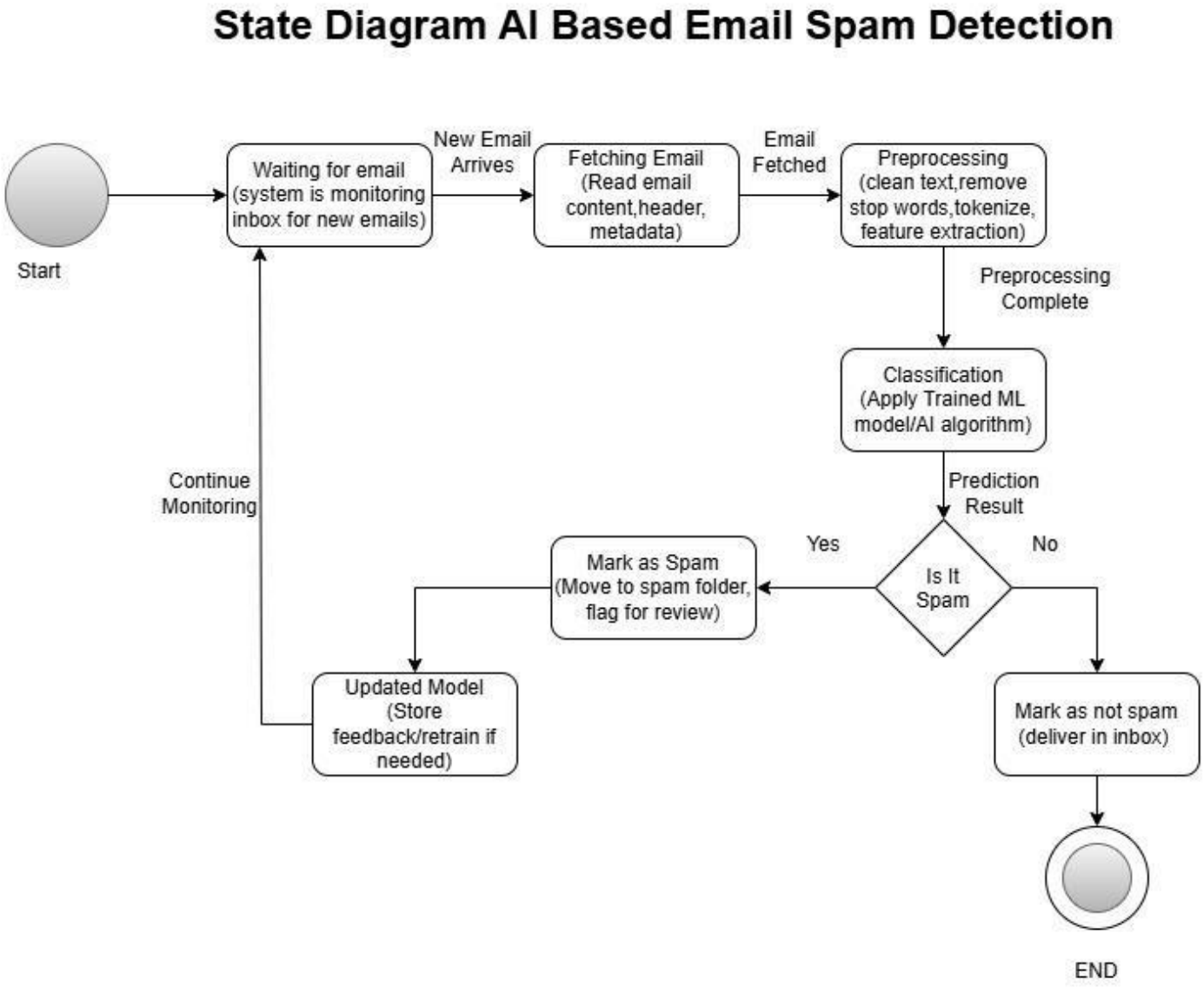
5. Sequence Diagram

5.1 Sequence Diagram



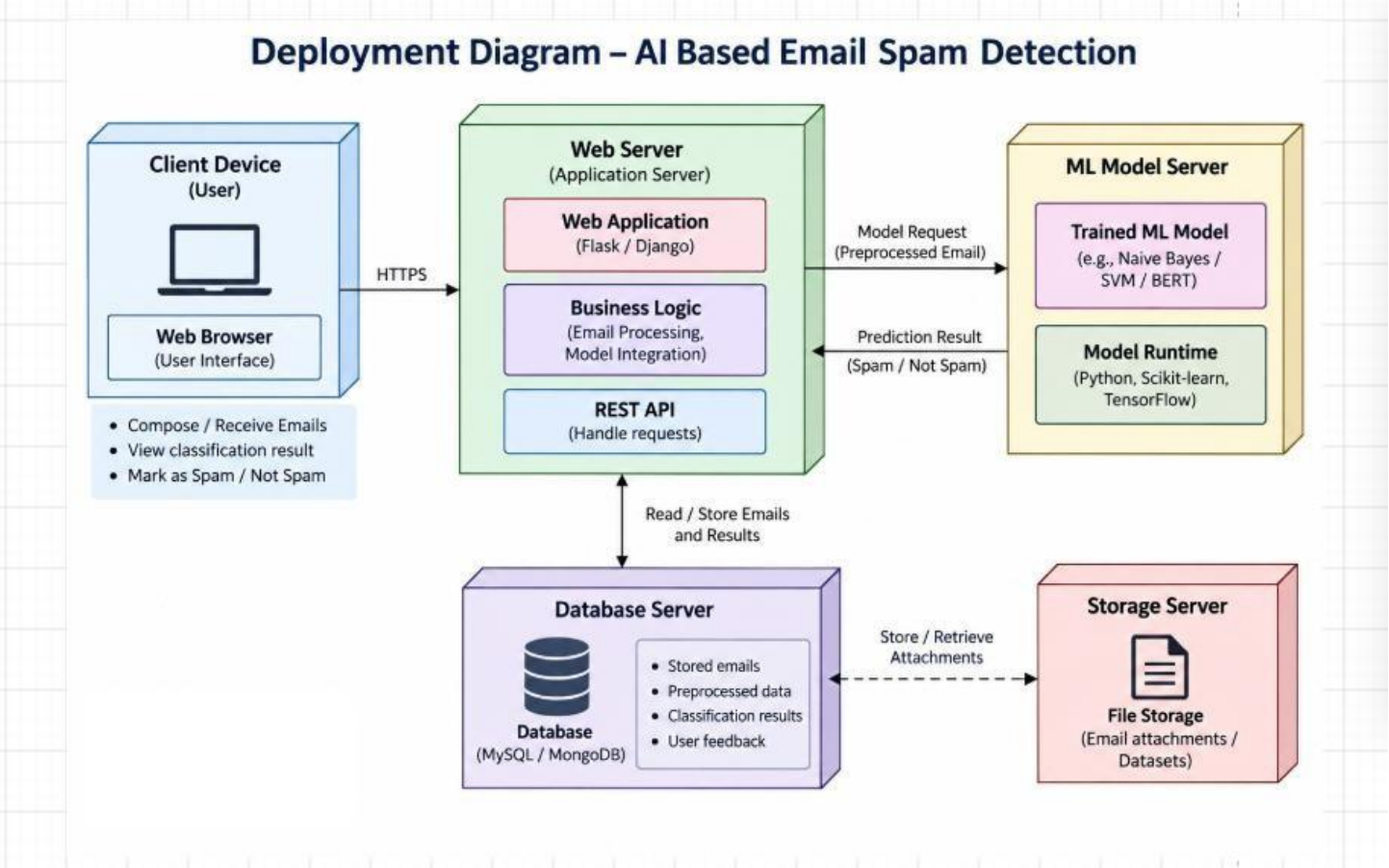
6. State Diagram

6.1 State Diagram



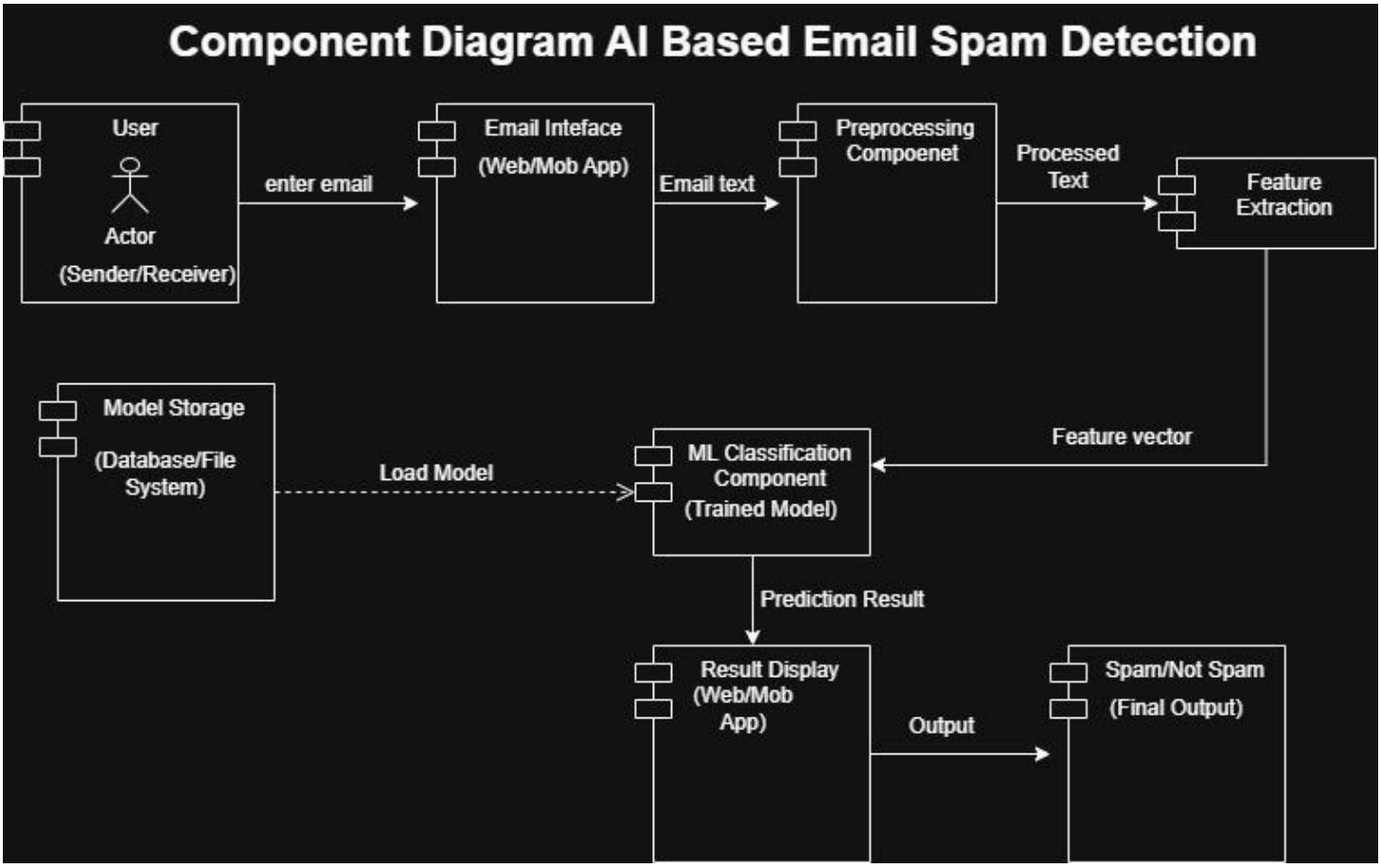
7. Deployment Diagram

7.1 Deployment Diagram



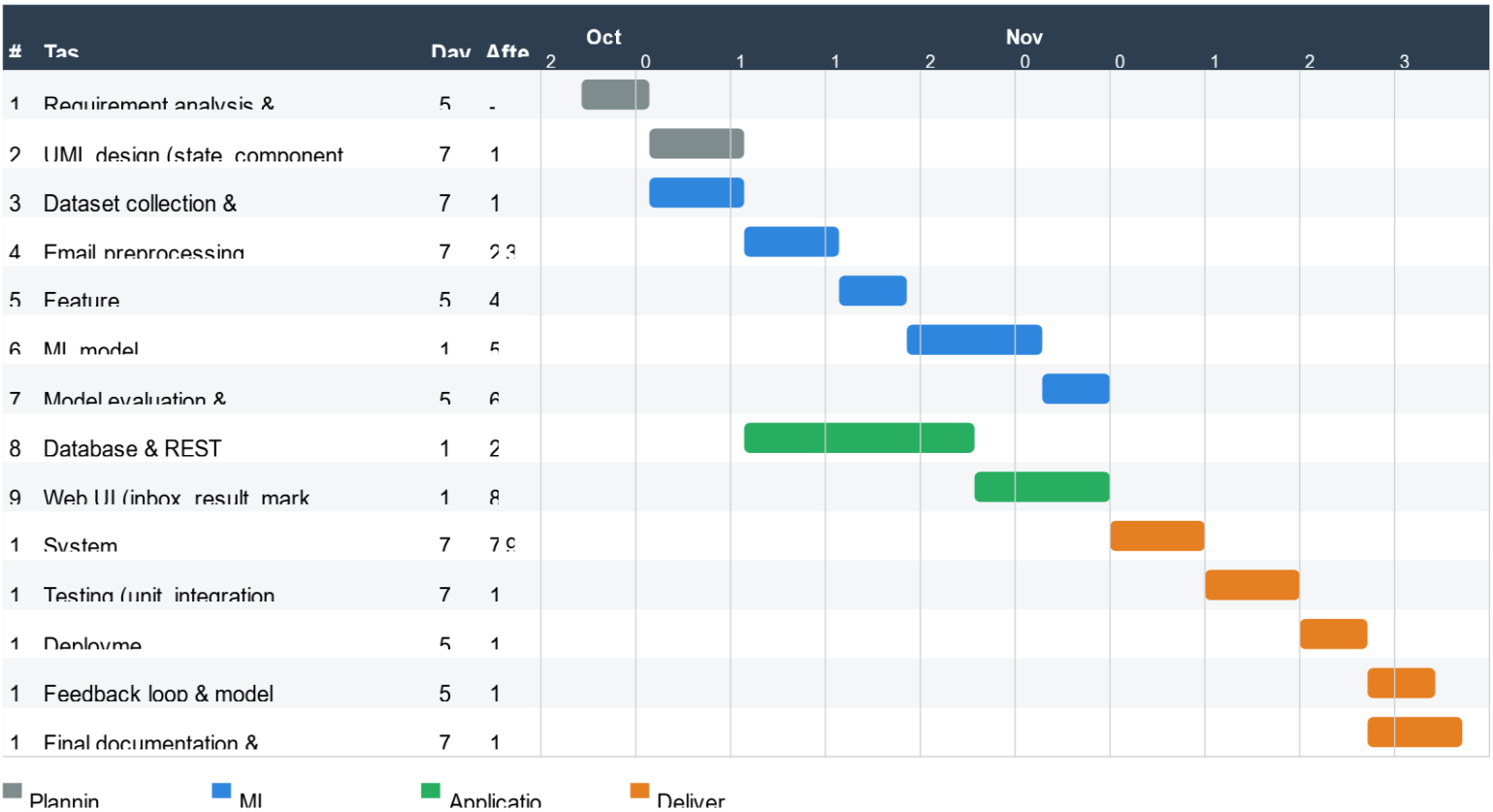
8. Component Diagram

8.1 Component Diagram



9. Gnatt Chart

9.1 Gnatt Chart Diagram



Project duration: 01 Oct 2026 - 04 Dec 2026 (65 days). 'After' = predecessor task

10. GitHub Link

10.1 GitHub Link

https://github.com/Aashi-Kothale/AI-Based-Spam-Detection-System